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PRN - B24CE1079

Subject - Object Oriented Programming

Assignment 9

/*PROBLEM STATEMENT: Implement a map associative container in which the keys will be the Permanent Registration Number (PRN) of the students and the values will be the name of the respective student. Develop a program that prompts the user for PRN and then it looks in the map, using the PRN as an index and returns the name of the student.*/

PROGRAM

```
#include <iostream>
#include <map>
#include <string>
using namespace std;
int main()
{
    typedef map <int,string> mapType;
    mapType studentMap;
    studentMap.insert(pair<int,string>(1234,"Suyash XYZ"));
    studentMap.insert(pair<int, string>(2222, "Sairaj"));
    studentMap.insert(pair<int, string>(3333, "Aary"));
    studentMap.insert(pair<int, string>(4444, "Anvi"));
    studentMap.insert(pair<int, string>(9999, "Srushti"));
    studentMap.insert(pair<int, string>(7777, "Vedant"));

    mapType::iterator iter;
    cout<<"=====Students of MMCOE=====\\n";
    cout<<"\\n Size of studentMap: "<<studentMap.size()<<"\\n";
    for (iter = studentMap.begin(); iter != studentMap.end(); iter++) {
        cout << iter->first <<": "<< iter->second<<"\\n";
    }

    int PRN;
    cout<<"\\n Enter PRN of the student :";
    cin>>PRN;
    iter = studentMap.find(PRN);
    if( iter!= studentMap.end() ){
        cout<<" "<<PRN <<" 's name is " <<iter->second ;
    }
    else{
        cout<<"Key is not in studentMap"<<"\\n";
    }
}
```

```

    }

    cout<<"\n";
    cout<<"\nErase PRN 3333\n";
    studentMap.erase(3333);
    cout<<"Map contains after erase\n";
    for (iter = studentMap.begin(); iter != studentMap.end(); iter++) {
        cout << iter->first <<": "<< iter->second<<"\n";
    }
    if(studentMap.empty()){
        cout<<"\nMap is empty\n";
    }
    else{
        cout<<"\nMap is not empty\n";
    }

    studentMap.clear();
    cout<<"\nSize of map after clear : "<<studentMap.size()<<"\n";
    return 0;
}

```

OUTPUT

```

=====Students of MMC0E=====

Size of studentMap: 6
1234: Suyash XYZ
2222: Sairaj
3333: Aary
4444: Anvi
7777: Vedant
9999: Srushti

Enter PRN of the student :4444
4444 's name is Anvi

Erase PRN 3333
Map contains after erase
1234: Suyash XYZ
2222: Sairaj
4444: Anvi
7777: Vedant
9999: Srushti

Map is not empty

Size of map after clear : 0

```

